Reg. No	R	A	2	4	1	1	0	2	6	0	2	0	0	9	1

B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2025

First & Second Semester

21CYB101J - CHEMISTRY

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer all Questions

		Marks	BL	CO
1.	The type of isomerism shown by the complex $[CoCl_2(en)_2]$ is ———. (A) Geometrical isomerism (B) Coordination isomerism (C) Linkage isomerism (D) Ionization isomerism	1	2	1
2.	In $K_4[Fe(CN)_6]$ the number of unpaired electrons in iron are? (A) 0 (B) 2 (C) 3 (D) 5	1	2	1
3.	The spin only magnetic moment value (in Bohr magneton units) of $Cr(CO)_6$ is——. (A) 0 (B) 2.84 (C) 4.90 (D) 5.92	1	2	1
4.	*Which is correct according to ligands in spectrochemical series?* (A) $I^- < Cl^- < H_2O <$en (B) $I^- < Cl^- < H_2O =$ en (C) $I^- = Cl^- < H_2O <$en (D) $I^- < Cl^- = H_2O <$en	1	1	1
5.	_____is the device used to measure the emf of the cell. (A) Voltmeter (B) Potentiometer (C) Ammeter (D) Multimeter	1	1	2
6.	The rusting of Iron is———. (A) Oxidation corrosion (B) Liquid metal corrosion (C) Wet corrosion (D) Corrosion by other gases	1	2	2
7.	Helmholtz free energy A is expressed as———. (A) A=U+TS (B) A=H+TS (C) A=U-TS (D) A=H-TS	1	1	2
8.	Passivation is due to formation of———. (A) higher EMF (B) lower EMF (C) metal oxide layer on metal (D) electrode potential	1	2	2
9.	Which of the following is least stable? (A) Anti conformation (B) Gauche conformation (C) Staggered conformation (D) Eclipsed conformation	1	2	3
10.	Which step in SN1 reaction is a slow rate determining step? (A) Attack of nucleophile (B) Formation of racemic mixture (C) Formation of transition state (D) Both a and b	1	2	3
11.	Which of the following is an initiator molecule in the free radical polymerisation? (A) Benzoyl peroxide (B) Sulphuric acid (C) Potassium permanganate (D) Chromium oxide	1	1	3

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			1	2	3
12.	Which of the following would exhibit co-ordination isomerism? (A) $[Cr(NH_3)_6][Co(CN)_6]$ (B) $[Co(en)_2Cl_2]^+$ (C) $[Cr(NH_3)_6]Cl_3$ (D) $[Cr(en)_2Cl_2]^+$		1	1	4
13.	Which among the following polymers have lowest solubility? (A) polyethylene (B) polystyrene (C) nylon 6 (D) epoxy resin		1	1	4
14.	What is the range of tensile strength, exhibited by fibres? (A) 300-3,000 (B) 4,000-15,000 (C) 20,000-150,000 (D) 5,000-10,000		1	1	4
15.	Which of the following is a co-polymer? (A) Polythene (B) Bakelite (C) PVC (D) Polyacrylonitrile		1	1	4
16.	Which of the following is NOT a natural polymer? (A) Rayon (B) Starch (C) Cellulose (D) RNA		1	1	4
17.	After the proportionality limit in the stress strain curve, we observe ..... (A) Ultimate point (B) Lower yield point (C) Upper yield point (D) Elastic limit		1	2	5
18.	In fiber reinforced composites which constituent will fail last ..... (A) Matrix (B) Filler (C) Both fail at same time (D) Cannot define		1	1	5
19.	Kevlar is a type of material. (A) Glass (B) Thermoplastic (C) Whisker (D) Polymer		1	1	5
20.	The source for XPS is —— (A) Mercury - arc (B) Nernst glower (C) Globar source (D) AlKα		1	1	5

PART - B (5 × 8 = 40 Marks)

Answer all Questions

		Marks	BL	CO
21.	(a) Describe the crystal field splitting in an octahedral complex with a neat diagram.	8	3	1
	(OR)			
	(b) Demonstrate, with proper examples, the isomerism exhibited in transition metal complexes.			
22.	(a) Derive the Nernst equation and discuss its applications.	8	3	2
	(OR)			
	(b) Explain in detail the mechanism with equations involved in dry corrosion.			
23.	(a) Sketch the potential energy diagram and explain in detail the conformational analysis of n-butane.	8	3	3
	(OR)			
	(b) Discuss in detail the SN^1 and SN^2 mechanisms with examples.			
24.	(a) Give differences between thermoplastic and thermosets.	8	3	4
	(OR)			
	(b) Write a short note on conducting polymer. Explain n-doping and p-doping in conducting polymers.			
25.	(a) Explain Bragg's Law with a neat diagram.	8	1	5
	(OR)			
	(b) Discuss on the ceramic and metal matrix composites with examples.			

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		Marks	BL	CO
	PART - C (1 × 15 = 15 Marks) **Answer any 1 Questions**			
26.	Give a neat sketch of the Pourbaix diagram and explain its significant features.	15	3	2
27.	Discuss on the principle and instrumentation of XPS. Give the merits and demerits of XPS analysis.	15	4	5

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